Data Cleanser Project – Explanation

Introduction

This project focuses on **data preprocessing**, which is an important step in any data science or machine learning workflow.
The goal is to clean a healthcare dataset by handling **missing values** and **outliers**, so that the data becomes reliable for analysis and modeling.

Dataset Description

The dataset contains patient health-related information such as:

- Age, Gender, Region
- BMI (Body Mass Index)
- Blood Pressure
- Cholesterol
- Glucose
- Disease Risk (target variable)

Some columns had missing values and some had extreme values (outliers), which needed to be handled.

Problem Statement

The dataset had two major issues:

1. **Missing Values**
    - High missing in gender
    - Moderate in region
    - Small in numerical columns like BMI, cholesterol, glucose
2. **Outliers**
    - Extreme values in BMI, cholesterol, and glucose
    - These can affect model accuracy

Approach Used

Step 1: Data Understanding

- Checked dataset structure using info() and describe()
- Calculated missing value percentage
- Identified problem areas

Step 2: Handling Missing Values

Three methods were applied:

1. Simple Imputer

- Numerical → Mean
- Categorical → Mode
- Easy but less accurate

2. KNN Imputer

- Uses nearest neighbors to fill missing values
- Works well for numerical patterns

3. MICE (Iterative Imputer)

- Predicts missing values using other features
- Maintains relationships between variables
- **Best method used in final dataset**

Step 3: Handling Outliers

Different techniques were applied:

1. Z-Score Method

- Detects values far from mean using standard deviation
- Replaced extreme values with median

2. IQR Method

- Uses quartiles (Q1, Q3)
- Caps values within range

3. Percentile Method

- Limits values between 1st and 99th percentile

4. Winsorization

- Replaces extreme values instead of removing them
- Keeps dataset size unchanged
- **Selected as final method**

Final Dataset

After applying:

- **MICE for missing values**
- **Winsorization for outliers**

The dataset became:

- No missing values
- Controlled outliers
- Balanced and usable

Results

- Improved data quality
- Preserved important patterns
- Prepared dataset for machine learning models

Conclusion

Data preprocessing is a critical step in data analysis.
This project shows how different techniques can be applied and compared to improve dataset quality.

The final cleaned dataset is now ready for:

- Machine learning
- Data analysis
- Visualization